Multimodal GenAI for Hospitality

A Project-II Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

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Under the Guidance of

Dr. Ritesh Joshi
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MEDICAPS UNIVERSITY

Department of Computer Science & Engineering
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MEDICAPS UNIVERSITY, INDORE- 453331

JAN-JUNE 2026

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Report Approval

The project work **“Multimodal GenAI for Hospitality”** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the “Project Report” only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner

Name:

Designation

Affiliation

Declaration

We hereby declare that the project entitled "**Multimodal GenAI for Hospitality"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology in 'Computer Science & Engineering' completed under the supervision of **Dr.Rtiesh Jhoshi, Prof. Vineeta Rathore, Prof. Pradeep Baniya** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

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Certificate

We, **Dr. Ritesh Jhoshi, Prof. Vineeta Rathore, Prof. Pradeep Baniya** certify that the project entitled **"Multimodal GenAI for Hospitality"**submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Vedang Rajoriya, Vandana Patel, Sumit Nigade, Suryapratap Sisodiya** is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Dr. Kailash Chandra Bandhu

Head of the Department

Computer Science & Engineering

Medicaps University, Indore

Acknowledgements

I would like to express my deepest gratitude to Honorable Chancellor, **Shri R C Mittal,** who has provided me with every facility to successfully carry out this project, and my profound indebtedness to **Prof. (Dr.) D. K. Patnaik,** Vice Chancellor, Medicaps University, whose unfailing support and enthusiasm has always boosted up my morale. I also thank **Prof. (Dr.) Ratnesh Litoriya,** Associate Dean of computing & Allied Branches, Medicaps University, for giving me a chance to work on this project. I would also like to thank my Head of Department **Dr. Kailash Chandra Bandhu** for his continuous encouragement for the betterment of the project.

I express my heartfelt gratitude to my External Guide, as well as to my Internal Guide without whose continuous help and support, this project would ever have reached to the completion.

I would also like to thank to my team at who extended their kind support and help towards the completion of this project.

Without their support this report would not have been possible.

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Abstract

This project presents the design and development of a **Multimodal and Multi-Agent AI-based Hospitality Experience Platform** that enhances the travel planning process through intelligent automation and immersive visualization. The system integrates a multi-agent architecture to generate structured travel itineraries, hotel recommendations, and budget estimates based on user inputs such as destination, duration, and preferences.

In addition, the platform utilizes generative AI techniques, including Large Language Models (LLMs) for producing descriptive narratives and text-to-image models for generating realistic visual representations of travel destinations. The backend is implemented using FastAPI to ensure efficient API orchestration, while the frontend is developed using modern web technologies to provide an interactive and visually engaging user experience.

The proposed system demonstrates the effective combination of Agentic AI and Generative AI to deliver personalized, real-time travel experiences. It reduces manual effort, improves decision-making, and provides users with a comprehensive preview of their journey. The modular and scalable design further allows future enhancements and integration with real-world travel services.

Keywords:

Multimodal AI, Agentic AI, Generative AI, Travel Planning System, Hospitality Platform, FastAPI, React, Text-to-Image Generation, Large Language Models, AI-based Recommendation System.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
GenAI	Generative Artificial Intelligence
LLM	Large Language Model
API	Application Programming Interface
UI	User Interface
UX	User Experience
DFD	Data Flow Diagram
ER	Entity Relationship
IDE	Integrated Development Environment
CI/CD	Continuous Integration / Continuous Deployment
AWS	Amazon Web Services
EC2	Elastic Compute Cloud
JSON	JavaScript Object Notation
GPU	Graphics Processing Unit
CPU	Central Processing Unit

Notations & Symbols

Symbol / Notation	**Meaning**
→	Flow of process / data
⇄	Two-way communication
%	Percentage
$	Cost / Budget representation
#	Identifier / Reference
[]	List / Array
{}	JSON Object / Structured Data
()	Function / Parameter grouping
API()	API function call
DB	Database
UI	User Interface layer

CHAPTER 1: INTRODUCTION

1.1 Introduction

The rapid evolution of Artificial Intelligence (AI) has transformed how users interact with digital systems, particularly in domains requiring personalization, automation, and visualization. The hospitality and travel industry, traditionally dependent on manual planning and fragmented platforms, has increasingly adopted AI-driven solutions to enhance user experience and decision-making.

This project presents a **Multimodal and Multi-Agent AI-based Hospitality Experience Platform** that integrates **Agentic AI** and **Generative AI** to deliver a comprehensive and immersive travel planning solution. The system enables users to input travel preferences such as destination, duration, budget, and interests, and in return generates a complete travel experience including itinerary planning, hotel recommendations, budget estimation, descriptive narratives, and AI-generated visual representations.

The core innovation lies in combining:

- A **multi-agent system** for intelligent planning and structured decision-making
- A **multimodal generative pipeline** for creating textual and visual outputs
- A **modern interactive frontend** for enhanced user engagement

By bridging the gap between planning and visualization, the system aims to provide users with not just information, but a **simulated experience of their travel journey before it begins**.

1.2 Literature Review

The integration of Artificial Intelligence in travel and hospitality has been explored through various approaches, including recommendation systems, chatbots, and itinerary planners. Traditional systems primarily rely on rule-based algorithms or collaborative filtering techniques, which often lack contextual understanding and personalization depth.

Recent advancements in **Large Language Models (LLMs)** have enabled systems to generate human-like text, making them suitable for dynamic itinerary generation and conversational interfaces. Studies in multimodal AI have further demonstrated the ability to combine text and image generation, enhancing user engagement through visual storytelling.

Research on **Agentic AI systems**, particularly multi-agent frameworks, highlights their effectiveness in decomposing complex tasks into specialized roles. For example, separating responsibilities such as data retrieval, planning, and summarization improves scalability and modularity.

However, existing solutions often face limitations:

- Lack of **integrated multimodal outputs**
- Limited **real-time personalization**
- Absence of **interactive and immersive user interfaces**
- Fragmented systems requiring multiple tools

This project addresses these gaps by developing a unified platform that combines:

- Multi-agent reasoning
- Generative AI capabilities
- Full-stack interactive design

1.3 Objectives

The primary objective of this project is to design and develop a **comprehensive AI-powered hospitality experience platform** that integrates planning and visualization capabilities.

Specific Objectives:

- To develop a **multi-agent system** capable of generating structured travel itineraries
- To integrate **LLM-based narrative generation** for immersive experience descriptions
- To implement **text-to-image generation models** for visualizing destinations and resorts
- To design a **scalable backend architecture** using FastAPI
- To build an **interactive and visually appealing frontend** using modern web technologies
- To ensure efficient API orchestration and data aggregation
- To optimize performance through asynchronous processing and caching mechanisms

1.4 Significance

The significance of this project lies in its ability to enhance the travel planning experience by combining intelligence, automation, and visualization.

Key Contributions:

- **User-Centric Experience**: Provides personalized travel plans tailored to user preferences
- **Multimodal Interaction**: Combines textual and visual outputs for better understanding
- **Automation**: Reduces manual effort in researching and organizing travel plans
- **Innovation in AI Integration**: Demonstrates the practical application of combining Agentic AI and Generative AI
- **Industry Relevance**: Aligns with current trends in AI-driven digital platforms

This system has potential applications in:

- Travel agencies
- Hospitality platforms
- AI-based recommendation systems
- Smart tourism solutions

1.5 Research Design

The research design adopted for this project is **applied and system-oriented**, focusing on the development and evaluation of a functional AI-based application.

Approach:

- **Design-Oriented Development**: Building a working system using modern AI technologies
- **Modular Architecture**: Separating components into agents, generative modules, and frontend
- **Iterative Development**: Continuous testing and refinement of system components

System Design:

The system is divided into three major layers:

1. **Agentic Layer**
    - Responsible for planning and structured output generation
    - Includes multiple agents such as Researcher, Writer, and Budget agents
2. **Generative Layer**
    - Handles narrative and image generation
    - Uses LLMs and text-to-image models
3. **Presentation Layer**
    - Frontend interface with interactive UI and 3D elements

Methodology:

- Input → Processing → Output model
- API-based integration
- Real-time data generation and rendering

1.6 Source of Data

The system primarily relies on **synthetic and AI-generated data**, supplemented by structured prompts and model outputs.

Data Sources:

- **User Inputs**:
    - Destination
    - Duration
    - Budget
    - Preferences

- **AI-Generated Data**:
    - Itinerary (multi-agent system)
    - Descriptions (LLM)
    - Images (text-to-image model)
- **Pre-trained Models**:
    - Language models for text generation
    - Image generation models for visualization

No external datasets are directly stored; instead, the system dynamically generates outputs based on input prompts and model capabilities.

1.7 Chapter Scheme

The report is organized as follows:

- **Chapter 1: Introduction**
  Provides an overview of the project, including objectives, significance, and research approach.
- **Chapter 2: Requirements Specification**
  Describes user characteristics, functional requirements, system dependencies, performance needs, and constraints.
- **Chapter 3: Design**
  Covers system design, algorithms (if applicable), and various diagrams such as DFD, UML, and database design.
- **Chapter 4: Implementation, Testing, and Maintenance**
  Explains technologies used, implementation details, testing methods, and user/installation guidelines.
- **Chapter 5: Results and Discussion**
  Presents system outputs, UI representation, module descriptions, and database snapshots.

- **Chapter 6: Summary and Conclusions**

  Summarizes the overall work and key findings.

- **Chapter 7: Future Scope**

  Discusses possible improvements and future enhancements.

CHAPTER 2: REQUIREMENTS SPECIFICATION

2.1 User Characteristics

The system is designed for a broad range of users with varying levels of technical expertise:

- **General Users (Travelers):**
  Individuals seeking personalized travel plans with minimal effort. Basic computer and internet knowledge is sufficient.
- **Students / Explorers:**
  Users interested in exploring destinations and experiences using AI-driven insights.
- **Technical Users / Developers:**
  Users who may analyze system outputs or integrate APIs; expected to have moderate technical knowledge.

The interface is designed to be intuitive, requiring no prior experience with AI systems.

2.2 Functional Requirements

The system must fulfill the following functionalities:

- Accept user inputs including destination, duration, budget, and preferences
- Generate a structured travel itinerary using a multi-agent system
- Provide hotel recommendations with key details
- Estimate overall travel budget
- Generate immersive textual descriptions using LLMs
- Produce AI-generated images based on user input
- Display results through an interactive frontend interface
- Handle API requests and responses efficiently
- Provide error handling and user feedback mechanisms

2.3 Dependencies

The system depends on the following external and internal components:

- **APIs:**
  - LLM API (for text generation)
  - Image generation API (for visual output)
- **Frameworks & Libraries:**
  - FastAPI (backend)
  - React, Three.js (frontend)
- **Environment:**
  - Internet connectivity for API calls
  - API keys for external services

2.4 Performance Requirements

- The system should respond within **2–5 seconds** under normal conditions
- Support concurrent user requests using asynchronous processing
- Efficient API handling to minimize latency
- Optimize resource usage through caching and modular design
- Ensure smooth frontend rendering and animations

2.5 Hardware Requirements

Minimum Requirements:

- Processor: Dual-core CPU
- RAM: 4 GB
- Storage: 500 MB free space

- Internet connectivity

Recommended Requirements:

- Processor: Quad-core CPU or higher
- RAM: 8 GB or more
- GPU (optional for enhanced rendering performance)

2.6 Constraints and Assumptions

Constraints:

- Dependency on external APIs for text and image generation
- API rate limits and usage costs
- Performance may vary based on network latency
- Limited control over third-party model outputs

Assumptions:

- Users have stable internet access
- API services remain available and functional
- Input data provided by users is valid and meaningful
- System is used in a standard web browser environment

CHAPTER 3: DESIGN

3.1 Algorithm

The system follows a structured workflow combining agent-based processing and generative AI:

Step 1: Accept user input (destination, days, budget, preferences)
Step 2: Process input through the multi-agent system:

- Researcher Agent → gathers relevant travel data
- Writer Agent → generates itinerary
- Budget Agent → estimates cost

Step 3: Generate narrative description using LLM
Step 4: Generate images using text-to-image model
Step 5: Aggregate all outputs into a structured format
Step 6: Return response to frontend for visualization

3.2 Function-Oriented Design

The system follows a modular functional approach:

- **Input Module:** Handles user input and validation
- **Agent Module:** Processes travel planning using multiple agents
- **Generative Module:** Handles text and image generation
- **Aggregator Module:** Combines all outputs
- **API Module:** Manages communication between frontend and backend
- **Presentation Module:** Displays results to users

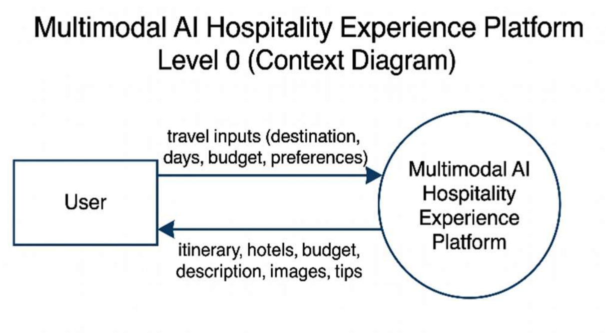
Each module operates independently, ensuring scalability and maintainability.

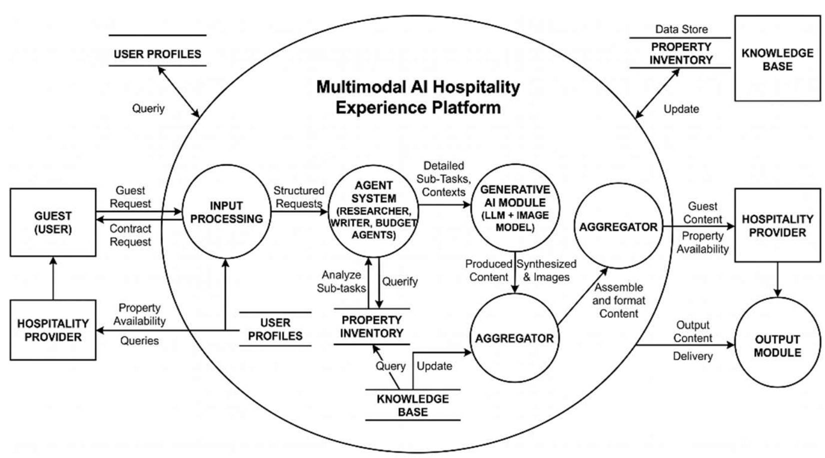
3.3 System Design

The system is designed using a layered architecture:

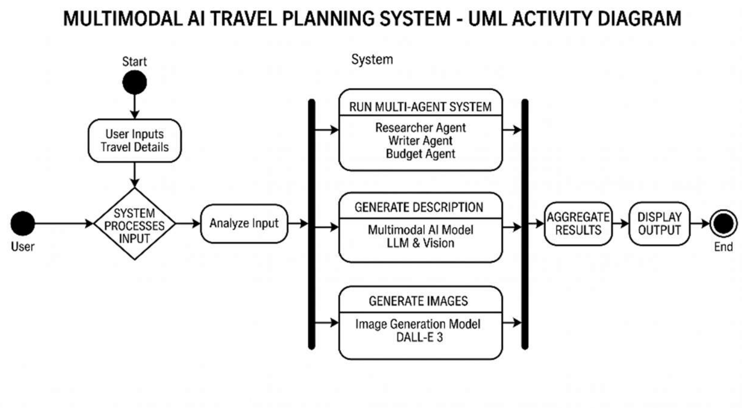
- **Frontend Layer:** User interface and interaction
- **Backend Layer:** API handling and orchestration
- **AI Processing Layer:** Agents and generative models
- **Integration Layer:** Aggregates outputs

3.3.1 Data Flow Diagrams

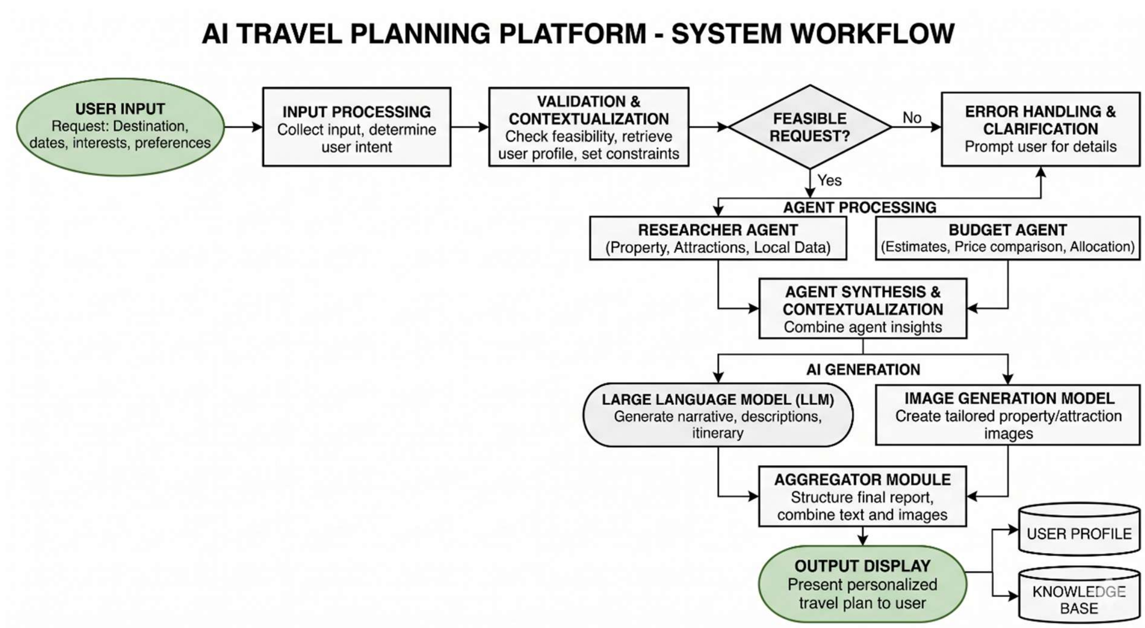




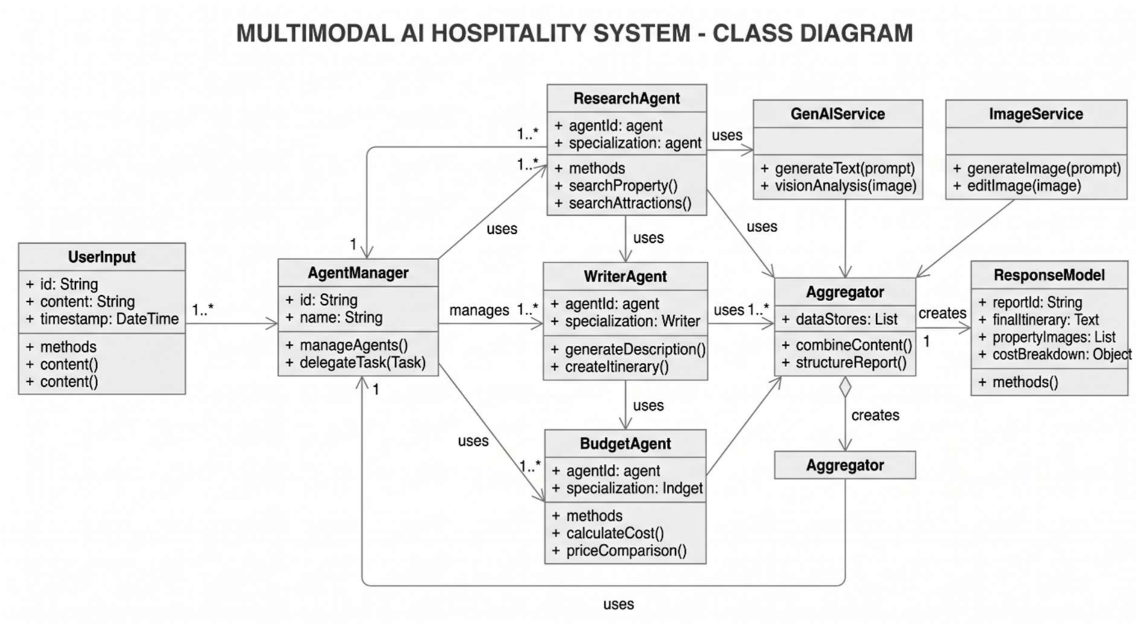
3.3.2 Activity Diagram



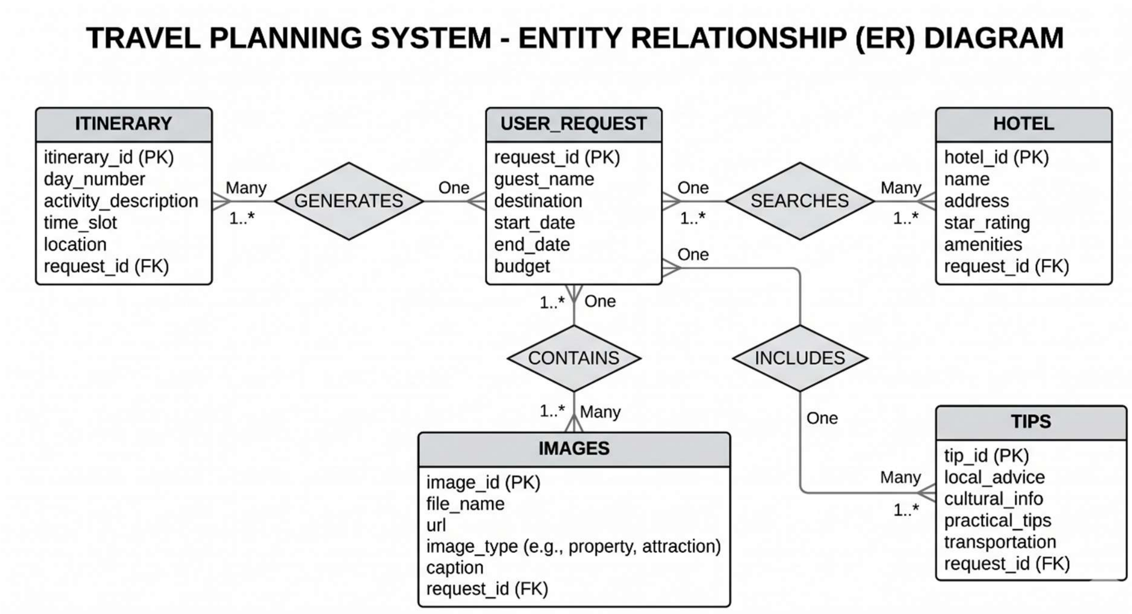
3.3.3 Flow Chart



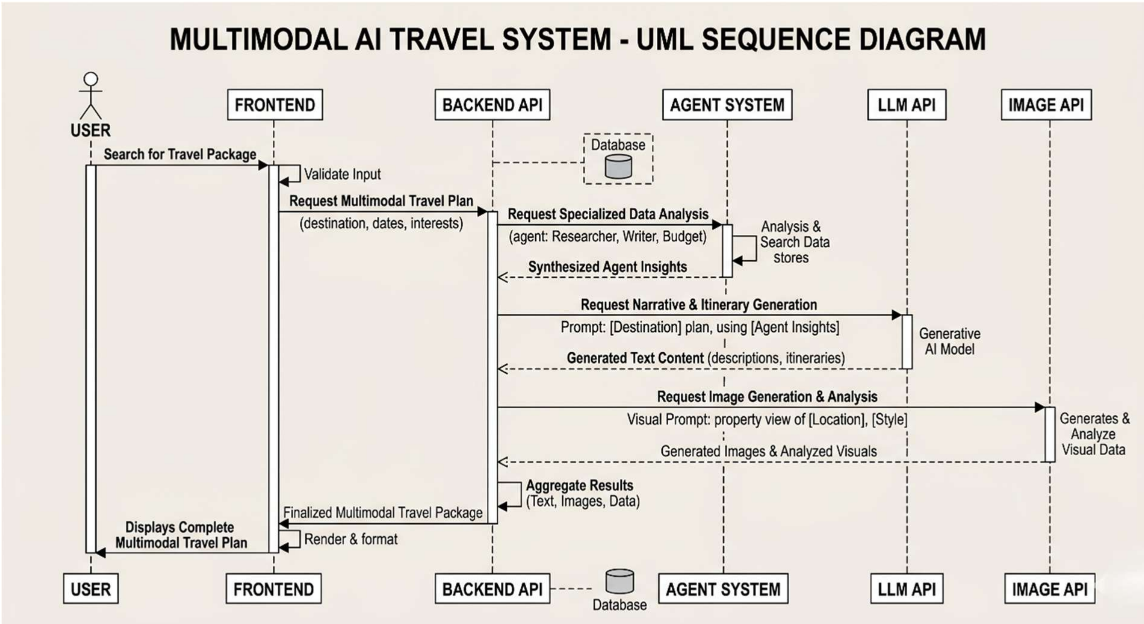
3.3.4 Class Diagram



3.3.5 ER Diagram



3.3.6 Sequence Diagram



CHAPTER 4: IMPLEMENTATION, TESTING, AND MAINTENANCE

4.1 Introduction to Languages, IDEs, Tools and Technologies

The system is implemented using modern web and AI technologies to ensure scalability, performance, and modularity.

Programming Languages:

- **Python** – used for backend development and AI integration
- **TypeScript / JavaScript** – used for frontend development

Frameworks and Libraries:

- **FastAPI** – backend framework for building REST APIs
- **React (Vite)** – frontend framework for building interactive UI
- **Tailwind CSS** – styling and responsive design
- **Framer Motion** – animations and transitions
- **React Three Fiber (Three.js)** – 3D visualization
- **Zustand** – state management

AI Technologies:

- **LLM API (Groq)** – for generating textual descriptions
- **Image Generation API (Flux via deAPI)** – for creating visuals

Development Tools:

- Visual Studio Code (IDE)
- Git & GitHub (version control)
- Postman (API testing)

4.2 Testing Techniques and Test Plans

The system is tested using multiple testing strategies to ensure reliability and correctness.

Testing Techniques:

- **Unit Testing:**
  Individual modules such as API endpoints and services are tested independently
- **Integration Testing:**
  Ensures proper interaction between agents, APIs, and frontend
- **Functional Testing:**
  Verifies that system outputs match user inputs and expected behavior
- **Performance Testing:**
  Evaluates system response time and API latency

Test Plan:

Test Case	Input	Expected Output
Valid Input	Destination, days, preferences	Complete itinerary and visuals
Invalid Input	Empty fields	Error message
API Failure	Simulated API error	Graceful fallback / error handling
High Load	Multiple requests	Stable performance

4.3 Installation Instructions

To set up the system, follow these steps:

Backend Setup:

1. Install Python (3.11 or above)
2. Clone the repository
3. Install dependencies using pip install -r requirements.txt
4. Configure environment variables (.env file)
5. Run the server using uvicorn main:app --reload

Frontend Setup:

1. Install Node.js
2. Navigate to frontend directory
3. Install dependencies using npm install
4. Start the development server using npm run dev

4.4 End User Instructions

1. Open the web application in a browser
2. Enter travel details:
    - Destination
    - Number of days
    - Budget
    - Preferences
3. Click on **"Generate Experience"**
4. Wait for the system to process inputs
5. View the generated:
    - Itinerary
    - Hotel suggestions
    - Budget estimate
    - AI-generated images
    - Experience description

The system provides a smooth and interactive experience with visual and textual outputs.

4.5 Maintenance

- Regular updates to API integrations
- Monitoring system performance and logs
- Updating dependencies and security patches
- Improving prompts and model outputs over time

4.5 Deployment Architecture and CI/CD Pipeline

The system is deployed using a cloud-based architecture to ensure scalability, reliability, and efficient performance. The deployment is carried out on Amazon Web Services using virtual servers and containerization.

Key Components:

- **Compute Service:**
  Amazon EC2 instances are used to host the backend and frontend services.
- **Containerization:**
  The application is containerized using Docker, ensuring consistency across development and production environments.
- **Container Orchestration:**
  Kubernetes is used to manage and scale containers efficiently.
- **Source Control:**
  Code is managed using GitHub.

4.5.1 Dockerization

The application is packaged into Docker containers to ensure portability and environment consistency.

Process:

- Backend and frontend are containerized separately
- Docker images are created using Dockerfiles
- Dependencies and runtime environments are encapsulated

Benefits:

- Simplified deployment
- Environment consistency
- Easy scaling and updates

4.5.2 CI/CD Pipeline

A Continuous Integration and Continuous Deployment (CI/CD) pipeline is implemented using GitHub Actions.

Workflow:

1. Code is pushed to the GitHub repository
2. Automated build process is triggered
3. Docker images are built and tested
4. Images are deployed to the server

Advantages:

- Automated deployment
- Reduced manual errors
- Faster development cycles

4.5.3 Kubernetes Deployment

Kubernetes is used to manage containerized applications.

Features:

- Automatic scaling of services
- Load balancing
- Self-healing of failed containers

Deployment Components:

- Pods (running containers)
- Services (network access)
- Deployments (scaling and updates)

4.5.4 Deployment Workflow

The overall deployment process is as follows:

1. Developer pushes code to repository
2. CI/CD pipeline builds Docker images
3. Images are deployed on EC2 instances
4. Kubernetes manages container orchestration
5. Application becomes accessible to users

4.5.5 Advantages of Deployment Architecture

- High scalability and availability
- Efficient resource utilization
- Faster deployment and updates
- Fault tolerance through orchestration
- Industry-standard DevOps practices

CHAPTER 5: RESULTS AND DISCUSSION

5.1 User Interface Representation

The system provides a modern and interactive user interface designed with a futuristic and user-friendly approach. The frontend includes a visually appealing layout with smooth animations and 3D elements to enhance user experience.

Key UI features include:

- Input form for travel details (destination, days, budget, preferences)
- Animated loading screen representing AI processing
- Dynamic dashboard displaying results
- Image gallery for AI-generated visuals
- Structured sections for itinerary, hotels, and budget

The interface ensures ease of use while maintaining a premium and immersive design.

5.2 Brief Description of Various Modules

The system is divided into the following core modules:

- **Input Module:**
  Collects and validates user inputs
- **Agent Module:**
  Generates itinerary and travel planning using multiple agents
- **Generative Module:**
  Produces textual descriptions and images using AI models
- **Aggregator Module:**
  Combines outputs from different modules into a unified response
- **Frontend Module:**
  Displays results in an interactive and visually rich format

Each module works independently, ensuring modularity and scalability.

5.3 Snapshots of System with Description

- **Homepage:**
  Displays input fields and 3D background interface
- **Loading Screen:**
  Shows animated AI processing steps
- **Output Dashboard:**
  Displays generated itinerary, hotels, and budget

- **Image Gallery:**
  Shows AI-generated destination visuals

Each screen demonstrates smooth transitions and real-time data rendering.

5.4 Backend Representation

The backend is developed using FastAPI and follows a modular architecture.

Key components include:

- API endpoints for handling requests
- Multi-agent system for planning
- Integration with LLM and image generation APIs
- Aggregation layer for combining outputs

Data is processed dynamically and returned in structured JSON format.

5.5 Database Representation

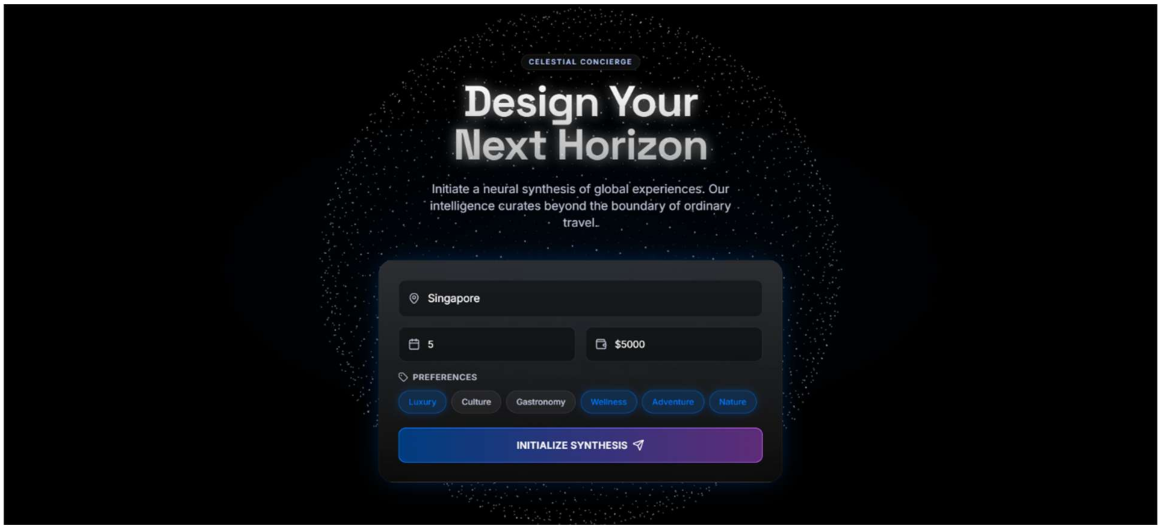
The system primarily uses dynamic data generation; however, optional storage can be implemented.

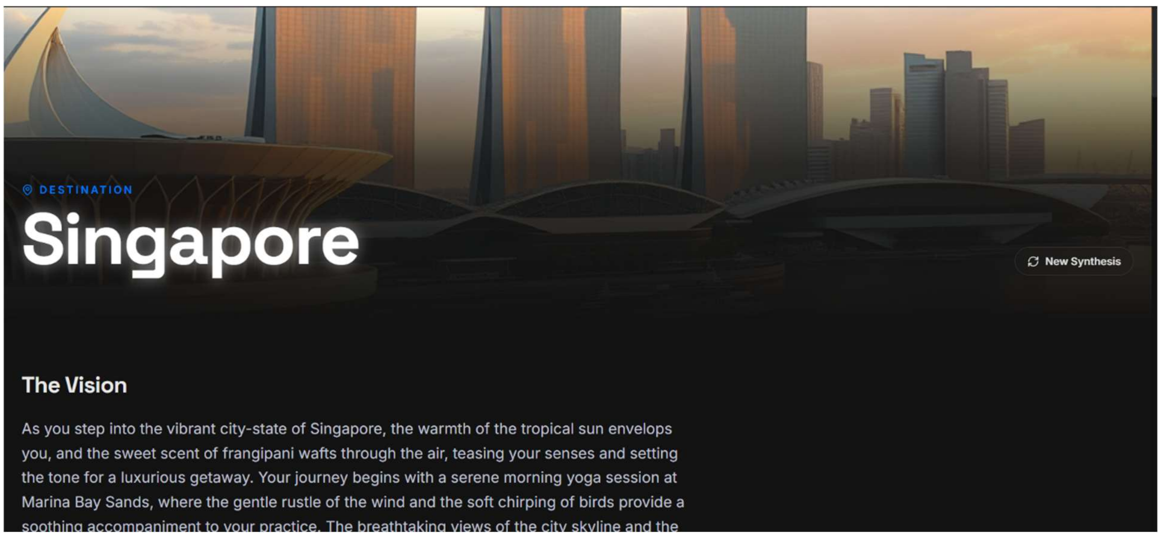
Database Structure:

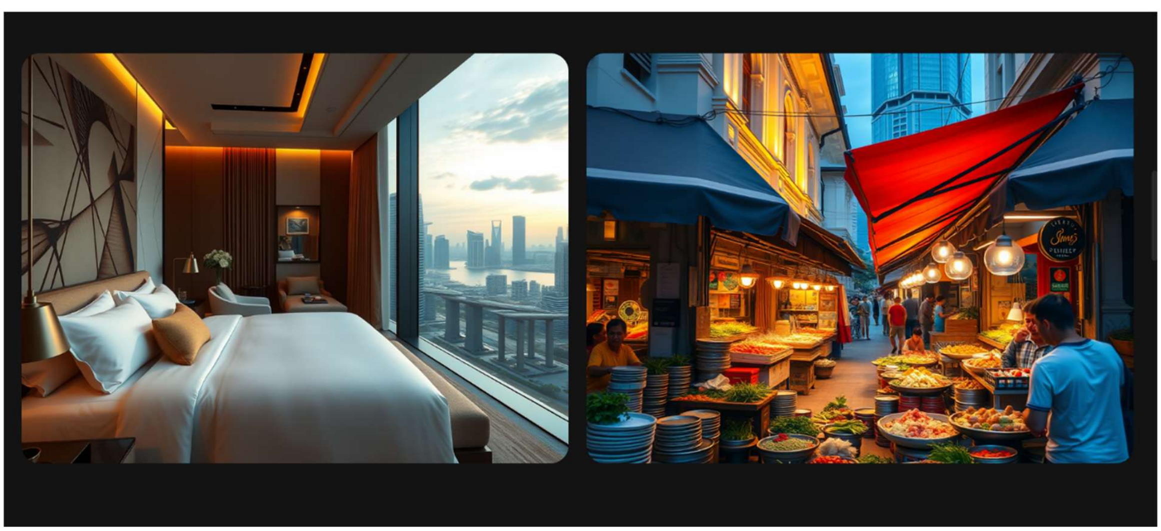
- User Input data
- Generated itinerary
- Hotel details
- Image URLs

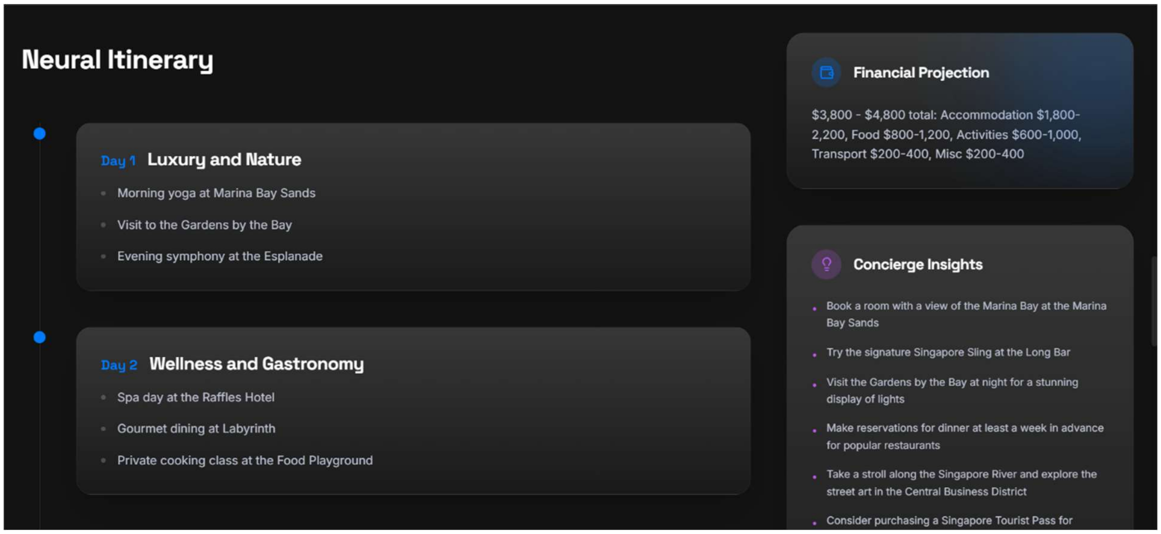
Description:

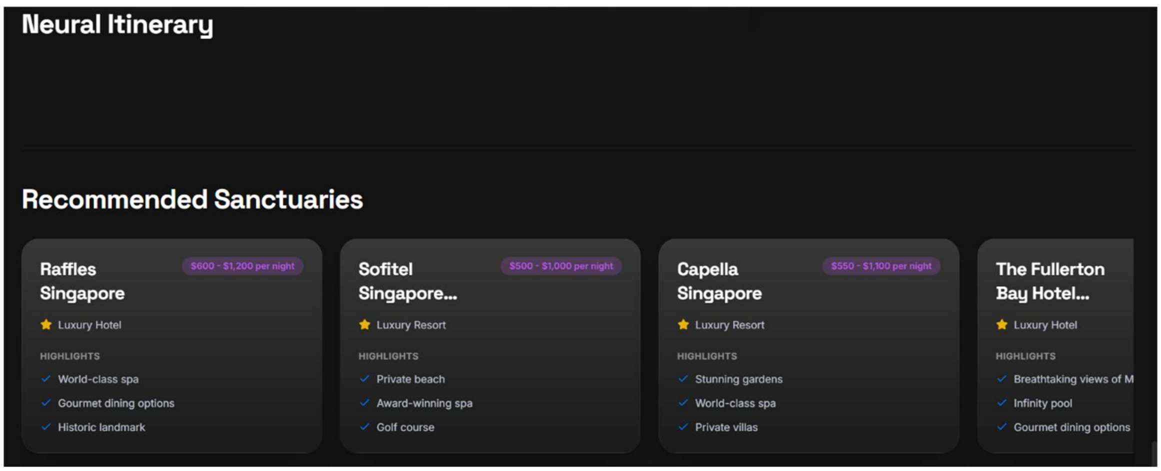
- Data is handled in JSON format
- No heavy relational database is required
- Optional caching or NoSQL storage can be used for performance


CELESTIAL CONCIERGE
Design Your Next Horizon
Initiate a neural synthesis of global experiences. Our intelligence curates beyond the boundary of ordinary travel.
Singapore
5
$5000
PREFERENCES
Luxury
Culture
Gastronomy
Wellness
Adventure
Nature
INITIALIZE SYNTHESIS


DESTINATION
Singapore
New Synthesis
The Vision
As you step into the vibrant city-state of Singapore, the warmth of the tropical sun envelops you, and the sweet scent of frangipani wafts through the air, teasing your senses and setting the tone for a luxurious getaway. Your journey begins with a serene morning yoga session at Marina Bay Sands, where the gentle rustle of the wind and the soft chirping of birds provide a soothing accompaniment to your practice. The breathtaking views of the city skyline and the







Discussion

The system successfully demonstrates the integration of Agentic AI and Generative AI to create a complete travel experience platform. The outputs are personalized, visually rich, and generated in real-time.

Key observations:

- The system provides accurate and structured travel plans
- Visual outputs enhance user understanding and engagement
- Performance is efficient due to asynchronous processing
- The modular design allows easy scalability and improvements

Overall, the system achieves its objective of delivering an intelligent and immersive hospitality experience.

CHAPTER 6: SUMMARY AND CONCLUSIONS

6.1 Summary

This project presents the development of a **Multimodal and Multi-Agent AI-based Hospitality Experience Platform**, designed to enhance the travel planning process through automation, personalization, and visualization.

The system integrates:

- A **multi-agent architecture** for intelligent itinerary planning
- **Large Language Models (LLMs)** for generating immersive textual descriptions
- **Text-to-image models** for visualizing travel destinations
- A **modern frontend interface** for interactive user experience

The backend, built using FastAPI, efficiently orchestrates multiple AI components and delivers structured outputs in real-time. The frontend, developed using React and advanced animation libraries, provides a visually rich and user-friendly interface.

The project successfully demonstrates how combining **Agentic AI and Generative AI** can create a unified and intelligent system capable of delivering complete travel experiences.

6.2 Conclusions

The developed system achieves its primary objective of simplifying and enhancing travel planning by providing:

- Personalized and structured itineraries
- AI-generated visual representations of destinations
- Immersive experience descriptions
- Efficient and responsive system performance

The modular architecture ensures scalability, maintainability, and ease of integration with future technologies. The project also highlights the practical applicability of advanced AI techniques in real-world domains such as hospitality and tourism.

CHAPTER 7: FUTURE SCOPE

The current system provides a strong foundation, with several opportunities for further enhancement:

- **Integration with Real-Time APIs:**
  Incorporating live data for flights, hotels, and pricing
- **User Authentication and Personalization:**
  Storing user history and preferences for improved recommendations
- **Advanced Agent System:**
  Adding more specialized agents such as booking, recommendation, and optimization agents
- **Enhanced Visualization:**
  Incorporating AR/VR-based travel previews
- **Mobile Application Development:**
  Extending the platform to Android and iOS applications
- **Database Integration:**
  Implementing persistent storage for user data and generated results
- **Performance Optimization:**
  Improving response time and scalability for large-scale deployment
- **Multilingual Support:**
  Enabling the system to support multiple languages for global users

APPENDIX

The appendix includes supporting materials related to system implementation, API structure, and deployment configuration.

A. Sample Input


```
{
  "destination": "Bali",
  "days": 5,
  "budget": "luxury",
  "preferences": ["beach", "spa", "sunset"]
}
```


B. Sample Output

The system generates a structured response including:

- Itinerary
- Hotel recommendations
- Budget estimate
- Experience description
- AI-generated images
- Travel tips

C. API Endpoint

POST /generate-experience

D. Technologies Used

- Backend: FastAPI
- Frontend: React, Tailwind CSS, Three.js
- AI Models: LLM (Groq), Image Generation (Flux)
- Tools: Git, Postman, VS Code

E. Deployment Configuration

The system is deployed using a cloud-native approach:

- Cloud Platform: Amazon Web Services (AWS)
- Compute: Amazon EC2
- Containerization: Docker
- Orchestration: Kubernetes

BIBLIOGRAPHY

The following references were used during the development of this project:

1. FastAPI Documentation – https://fastapi.tiangolo.com/
2. React Documentation – https://react.dev/
3. Tailwind CSS – https://tailwindcss.com/
4. Three.js – https://threejs.org/
5. Framer Motion – https://www.framer.com/motion/
6. Docker Documentation – https://www.docker.com/
7. Kubernetes Documentation – https://kubernetes.io/
8. GitHub Actions Documentation – https://docs.github.com/actions
9. Amazon EC2 – https://aws.amazon.com/ec2/
10. Artificial Intelligence and Machine Learning resources